UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester I 2026/2027

Subject	:	Software Engineering (SECP3204)
Task	:	**Academic Paper in Literature Review**
Due	:	4/11/2026 (Wednesday)
Submission	:	Upload softcopy short paper (in .pdf format) via e-learning.
File name	:	Group number
Type	:	Group (3 students per group)

	Topics
1	Evaluating Software Engineering Practices in Student Data System Projects
2	The Impact of Database Design on Software Quality in Data-Driven Applications
3	The Role of Generative AI in Learning Software Engineering Among Data Engineering Students
4	Applying Agile Software Development Practices in Undergraduate Data Engineering Projects
5	Integrating Software Engineering and Data Engineering Principles in Data Pipeline Development
6	Requirements Engineering Challenges in Developing Real-World Information Systems
7	Evaluating Software Testing Practices in Student-Developed Web Applications
8	Consistency Between System Requirements, UML Models, and Software Implementation
9	The Relationship Between User Interface Usability and User Satisfaction in Data-Driven Applications
10	The Effect of Input Validation and Error Handling on the Reliability of Web-Based Information Systems
11	The Relationship Between Version Control Practices and Team Collaboration in Student Software Projects

Instruction:

1. Example of academic phase (https://www.phrasebank.manchester.ac.uk/).
2. Students are required to prepare **IEEE SHORT PAPER**. Refer to the provided IEEE paper format for your reference. The short paper should be written using the provided IEEE paper template (ieee format.pdf).
3. The length of this short paper **SHOULD NOT EXCEED FOUR (4) PAGES**.
4. The short paper should include abstract, keywords, introduction, main body (content) conclusion and references. Create your own suitable title for your short paper. Moreover, you should format your short paper according to the given template for IEEE paper.
5. The content of the short paper should include, depending on the topic, use your creativity to design the content.
6. Learn how to cite references using UTM (Harvard style such as author, year) and you have to cite at least seven (7) references.
7. Log book and process to collect the research material, record all the processes to produce a paper such as the keywords used in search engines or the prompt in Ai generate. (Example of recording the date, student name, activity). All the material should be recorded in GitHub or log book. Write all the prompt use in any GAi.
8. **PLAGIARISM** of the content is prohibited. If any of the content is known to be copied from other websites/reviews/articles/blogs, you will be given a ZERO **(0) mark**. Include the Turnitin report.
9. You can get photos/information from other resources, but you must **include credits (in citation/link)** to the original owners.

1. Assessment structure

Suggested total: 100 marks

Assessment component	Weight
A. Academic paper/report	60%
B. Oral presentation	30%
C. Question-and-answer session	10%
Total	100%

2. Rubric A: Academic paper/report (60 marks)

Use the following criteria to assess the students' research reports.

Criteria	Excellent	Good	Satisfactory	Needs improvement	Marks
1. Introduction and problem statement	Problem is clear, relevant, and well justified.	Problem is clear with reasonable justification.	Problem is identified but lacks justification.	Problem is unclear or irrelevant.	10
2. Research objectives and questions	Clear, measurable, and aligned with the problem.	Mostly clear and aligned.	Partially aligned or too broad.	Unclear or missing.	5
3. Literature review	Uses relevant academic sources and critically compares previous studies.	Uses relevant sources with some comparison.	Limited sources and mostly descriptive.	Insufficient or unreliable sources.	10
4. Research methodology	Appropriate method, data collection, and analysis are clearly explained.	Method is appropriate but some details are missing.	Method is described but weakly justified.	Method is inappropriate or unclear.	10
5. Results and discussion	Findings are clearly presented, analysed, and linked to research questions and literature.	Findings are presented with reasonable analysis.	Limited analysis or interpretation.	Findings are unclear or unsupported.	10
6. Conclusion and recommendations	Answers the research questions and provides practical	Summarises findings with relevant recommendations.	Basic summary with limited recommendations.	Conclusion is unsupported or incomplete.	5

	recommendations.				
7. Academic writing and references	Well structured, professional, properly cited, and consistent in referencing.	Generally clear with minor errors.	Several writing or referencing errors.	Poor structure, citation, or academic integrity.	10
Total					60

3. Rubric B: Oral presentation (30 marks)

Criteria	Excellent	Good	Satisfactory	Needs improvement	Marks
1. Content and organisation	Logical flow covering problem, objectives, method, findings, and conclusion.	Mostly logical and complete.	Some important content is missing.	Disorganised or incomplete.	8
2. Understanding of research	Demonstrates strong understanding of the topic and findings.	Demonstrates good understanding.	Demonstrates basic understanding.	Shows limited understanding.	8
3. Visual aids and slides	Clear, professional slides with effective diagrams, tables, and visualisations.	Good slides with minor issues.	Slides are readable but text-heavy or inconsistent.	Slides are unclear or poorly designed.	5
4. Communication and delivery	Clear, confident, engaging, and within the allocated time.	Generally clear and within time.	Uneven delivery or minor timing issues.	Difficult to follow or substantially over time.	5
5. Team participation	All members contribute appropriately and demonstrate preparation.	Most members contribute effectively.	Uneven contribution.	One or more members contribute very little.	4
Total					30

4. Rubric C: Question-and-answer session (10 marks)

Criteria	Excellent	Good	Satisfactory	Needs improvement	Marks
Response accuracy and relevance	Answers accurately with clear justification and evidence.	Answers correctly with reasonable explanation.	Gives partially correct or general answers.	Cannot answer or gives incorrect answers.	5
Critical thinking	Evaluates limitations, explains decisions, and considers alternatives.	Explains decisions and some limitations.	Provides limited reasoning.	Shows little critical thinking.	5
Total					10

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6. Recommended report structure for students (suggested)

To align the report with the rubric, ask students to prepare the following sections:

1. Title and abstract
2. Introduction and problem statement
3. Research objectives and questions
4. Literature review
5. Research methodology
6. Results and discussion
7. Conclusion and recommendations
8. References
9. Appendices, where applicable

Academic integrity note: Require students to cite all sources correctly, verify references, and disclose the use of generative AI according to the course guidelines. AI-generated content should not substitute for students' own analysis or fabricated research findings.